

Quiz 1 answers

1. (a) scalar multiple

(b) $2022 e_1 + e_2 = (2022 - \frac{1}{r}) e_1 + \frac{1}{r} (e_1 + r e_2)$ Answer: $\begin{bmatrix} 2022 - \frac{1}{r} \\ \frac{1}{r} \end{bmatrix}$

2. blank: $A(x-y)=0$

3. (a) The three nonzero rows of

the matrix on the board, which is in row echelon form (not reduced REF but that doesn't matter)

I $\Leftrightarrow$ Rows indep (None)

(None) Rows span $\mathbb{R}^c \Leftrightarrow$ II

(None) Cols indep $\Leftrightarrow$ II

I $\Leftrightarrow$ Cols span $\mathbb{R}^r$ (None)

I $\Leftrightarrow$ Pivot in each row (None)

(None) Pivot in each col $\Leftrightarrow$ II

I $\Rightarrow r \leq c$ (None)

(None) $c \leq r \Leftarrow$ I

(b) $\{v_1, v_3, v_4\}$ pivot cols of given matrix

(c) $\dim \text{Nul}(M) = 2$ A basis: $\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -17 \\ 0 \\ 11 \\ -7 \\ 1 \end{bmatrix}$

4. (a) $E_{ij} E_{kl} = \delta_{jk} E_{il}$

(b) $\left\{ \begin{bmatrix} a & b \\ 0 & a \end{bmatrix} \mid a, b \in \mathbb{R} \right\}$

(c) $M =$ any solution of a homogeneous system of n^2 equations (equalities for n^2 entries of AM and MA) in n^2 variables (entries of M). We can solve this in the usual way by row reduction. Solutions will form a vector space (e.g. in (b) the dim of solution space is 2).

5. (a) $\begin{bmatrix} \frac{1}{r} & 0 \\ 0 & \frac{1}{r+1} \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -r \\ 0 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 2021 & 2022+r \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 - 2021R_2} \begin{bmatrix} 1 & 0 & 1+r \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[R_1 - (1+r)R_3]{R_2 - R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{same}} \begin{bmatrix} 1 & -2021 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{same}} \begin{bmatrix} 1 & -2021 & -(1+r) \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \text{Answer}$

HW2 answers

2.2 + 3.1) (a) yes, $\{E_{ii}, E_{ij} + E_{ji} (i \neq j)\}$ is one basis

(b) No (O is not invertible) (c) Yes

3.3) Let $ax^2 + b \cos x + ce^x = 0$ function for $a, b, c \in \mathbb{R}$, i.e. equation is true for each real value of x .

Substitute suitable values for x to deduce a, b, c must all be 0 by solving a system of eqⁿ.

You can play other games e.g. take $\lim_{x \rightarrow -\infty}$ (why valid) to first get $a=0$, then $b=0$ and hence $c=0$ (how?)

3.7) Let $e_1, \dots, e_m$ be the standard basis of $\mathbb{R}^m$ and

let $f_1, \dots, f_n$ be the standard basis of $\mathbb{R}^n$.

See that $e_i f_j^t = E_{ij}$ (matrix units) and these form a basis of $\mathbb{R}^{m \times n}$.

Fix i and j . Then $e_i = \sum_{k=1}^m a_k X_k$ (as X_k span $\mathbb{R}^m$) and
 $(1 \leq i \leq m, 1 \leq j \leq n)$ $f_j = \sum_{l=1}^n b_l Y_l$ (as Y_l span $\mathbb{R}^n$)

Now $e_i f_j^t = \left(\sum_k a_k X_k \right) \left(\sum_{l=1}^n b_l Y_l \right)^t = \sum_{k,l} a_k b_l X_k Y_l^t$ is in the span of the given set $X_k Y_l^t$

So $X_i Y_j^t$ is a set of mn vectors that span a v-space of dim mn . Thus the given set is a basis.

4.1) (a) Show $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 1 \\ 0 & 2 & 1 \end{bmatrix} \xrightarrow[\text{reduction}]{\text{row}} \text{Id}$ (b) Solve $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

4.3) Sketch: write $B' = (w_1, w_2, \dots, w_n) =$ arbitrary basis. Write

$$w_i = a_{i1}v_1 + a_{i2}v_2 + \dots + a_{in}v_n$$

Matrix $A = (a_{ij})$ is an invertible matrix. Symbolically in terms of hypervectors we have

$$[w_1 \dots w_n] = [v_1 \dots v_n] A$$

Given operations (i), (ii), (iii) amount to using $A =$ an elementary matrix

As any invertible A can be written as a product of elementary matrices, the result follows.

6) $A \xrightarrow[\text{reduce}]{\text{row}} \begin{bmatrix} 1 & 2 & 0 & 13 & 7/2 \\ 0 & 0 & 1 & -3 & 1/2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ Row operations preserve row space and do not change (in)dependence of columns.
 $\therefore$ Basis of row space of $A =$ the two nonzero rows of R
 Basis of col space of $A =$ 1st and 3rd columns of A (cols of R with pivots)

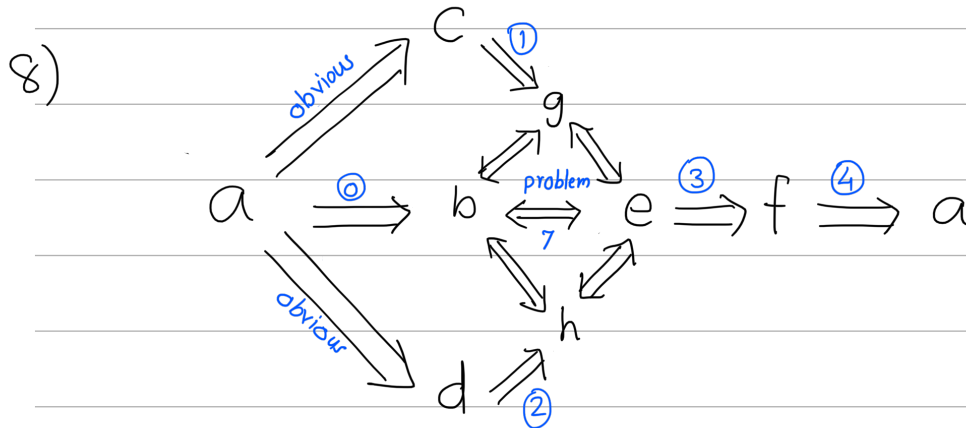
$Ax=0 \Leftrightarrow Rx=0$. General solution = $x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -13 \\ 0 \\ 3 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -7/2 \\ 0 \\ -1/2 \\ 0 \\ 1 \end{bmatrix}$ These three vectors are indep. (see slots 2,4,5) $\Rightarrow$ a basis of null space

7) $\text{REF}(A)$ has at most one pivot in each row and at most one pivot in each col.

From (ii) $\Leftrightarrow \text{RREF}(A)$ has a pivot in each column (i.e. no free variables)

table (iii) $\Leftrightarrow \text{RREF}(A)$ has a pivot in each row

Thus any two of i,ii,iii imply $\text{RREF}(A)=I_d$ which implies all three statements



⊙ If A is invertible, $x \mapsto A^{-1}x$ is the inverse function of $x \mapsto Ax$.

① Assume (c). If $Ax=0$ then $LAx=0$ i.e. $Ix=x=0$. So $x \mapsto Ax$ is injective. Use problem 7.

② Assume (d). Then $ARb=Ib=b$. So $x=Rb$ solves $Ax=b$. Thus $x \mapsto Ax$ is surjective. Use problem 7.

③ Assume (e). Then $E_k \cdots E_1 A = I$, where $E = E_k \cdots E_1$ is a product of elementary matrices E_i .

Therefore $A = E_1^{-1} \cdots E_k^{-1}$ because E_i are invertible. Each E_i^{-1} is also an elementary matrix (encoding the reverse of the row operation corresponding to E_i), giving (f).

④ Product of invertible matrices is invertible: $(XY)^{-1} = Y^{-1}X^{-1}$ by inspection

HW 3 answers

4.1c + 4.2ab) straightforward $B' = BP$ e.g. in 4.2a $[e_1 + e_2, e_1 - e_2] = [e_1, e_2] \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

say i^{th} row $= [a_1 \dots a_n]$

1.4) There is exactly one linearly indep row₂ (equivalently, column) and all others

are scalar multiples of it. Then $A = \begin{bmatrix} \lambda_1 a_1 & \dots & \lambda_1 a_n \\ \lambda_2 a_1 & \dots & \lambda_2 a_n \\ \vdots & & \vdots \\ \lambda_m a_1 & \dots & \lambda_m a_n \end{bmatrix} = \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_m \end{bmatrix} \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix}$ Both vectors can be scaled by nonzero scalars reciprocal to each other.

$$2.1) \quad A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \sum a_{ij} e_{ij} \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \sum b_{ij} e_{ij}$$

M

AMB

First column of desired matrix

$$e_{11} \quad \begin{bmatrix} a_{11} b_{11} & a_{11} b_{12} \\ a_{21} b_{11} & a_{21} b_{12} \end{bmatrix} = \underline{a_{11} b_{11}} e_{11} + \underline{a_{11} b_{12}} e_{12} + \underline{a_{21} b_{11}} e_{21} + \underline{a_{21} b_{12}} e_{22}$$

$$e_{pq} \quad A e_{pq} B = \underline{a_{1p} b_{q1}} e_{11} + \underline{a_{1p} b_{q2}} e_{12} + \underline{a_{2p} b_{q1}} e_{21} + \underline{a_{2p} b_{q2}} e_{22}$$

$$(a_{1p} e_{1p})(e_{pq})(b_{q1} e_{q1})$$

$$\text{Desired matrix} = \begin{array}{cccc|c} & e_{11} & e_{12} & e_{21} & e_{22} & \\ \begin{bmatrix} a_{11} b_{11} & a_{11} b_{12} & a_{12} b_{11} & a_{12} b_{21} \\ a_{11} b_{12} & a_{11} b_{22} & a_{12} b_{12} & a_{12} b_{22} \\ a_{21} b_{11} & a_{21} b_{21} & a_{22} b_{11} & a_{22} b_{21} \\ a_{21} b_{12} & a_{21} b_{22} & a_{22} b_{12} & a_{22} b_{22} \end{bmatrix} & e_{11} & e_{12} & e_{21} & e_{22} \end{array}$$

2.4) Sketch: Doing an elementary row (respectively, column) operation on A is equivalent to multiplying A on the left (respectively, right) by an elementary matrix. Thus it is enough to show that $A \xrightarrow[\text{col ops}]{\text{row and}} \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}$

Now $A \xrightarrow{\text{row op}} R = \text{RREF}(A)$. Now kill each nonpivot entry r_{ij} by using the pivot in i^{th} row:

if this $1 = r_{ik}$ (must have $k < j$), then do $\text{Col}_j - r_{ij} \text{Col}_k$. This operation does not change any other entry in R as $r_{ik} = 1$ is the only nonzero entry in Col_k of R.

Now switch columns to move all zero columns to the right, past all pivot cols.

3.1) There are two cases depending on parity of n .

- If $n=2k$ is even, $\ker T = \{(x_1, x_2, \dots, x_k, -x_k, \dots, -x_1)^t \mid x_i \in \mathbb{R}\}$ which has dimension $k = \frac{n}{2}$ (e.g. $e_1 - e_n, e_2 - e_{n-1}, \dots, e_k - e_{k+1}$ is a basis.)

Then by dimension formula $\dim(\operatorname{im} T) = n - k = k = \frac{n}{2}$

- If $n=2k+1$ is odd, $\ker T = \{(x_1, x_2, \dots, x_k, 0, -x_k, \dots, -x_1)^t \mid x_i \in \mathbb{R}\}$

$\dim(\ker T) = k = \frac{n-1}{2}$ ($e_1 - e_n, \dots, e_k - e_{k+2}$ being a basis.)

Then by dimension formula $\dim(\operatorname{im} T) = n - k = \frac{n+1}{2}$

3.3) Claim: There is a vector $v \in V$ such that $T(v) \neq cv$ for any scalar c .

Proof: If not, pick a basis v_1, v_2 and let $T(v_1) = c_1 v_1, T(v_2) = c_2 v_2$. If $c_1 = c_2 = c$ then for any $v \in V$, let $v = \alpha v_1 + \beta v_2$. $T(v) = T(\alpha v_1 + \beta v_2) = \alpha T(v_1) + \beta T(v_2) = c(\alpha v_1 + \beta v_2) = cv$.

This contradicts the hypothesis that $T \neq c \operatorname{Id}$. Thus $c_1 \neq c_2$.

Now $T(v_1 + v_2) = c_1 v_1 + c_2 v_2$, which is not a scalar multiple of $v_1 + v_2$. This proves the claim.

Pick v as in the claim. As $v \nmid Tv$ are lin indep and V has dimension 2, $\{v, Tv\}$ is a basis of V . To write the matrix of T wrt this basis, we write

$$\begin{cases} T(v) = 0 \cdot v + 1 \cdot T(v) \\ T(T(v)) = a \cdot v + b \cdot T(v) \end{cases}$$

$$\text{Desired matrix} = \begin{array}{cc|c} & v & T(v) & \\ \hline & 0 & a & v \\ & 1 & b & T(v) \end{array}$$

7) (a), (b) are straightforward e.g. (1) by definition $T_1 + T_2(v) := T_1(v) + T_2(v)$.

$T_1 + T_2$ is linear because $T_1 + T_2(cv + v') = T_1(cv + v') + T_2(cv + v')$ and now use linearity of T_1 & T_2 with commutativity of $+$ in W

(2) Addition & scalar multiplication in $\operatorname{Hom}(V, W)$ satisfy desired properties due to the same properties in W .

To show equality of functions $f=g$ means showing $f(x)=g(x)$ for each $x \in \text{domain}$

$$\begin{aligned} (c) \quad (S \circ (T_1 + T_2))(v) &= S((T_1 + T_2)(v)) \text{ by definition of } \circ \\ &= S(T_1(v) + T_2(v)) \text{ by definition of } T_1 + T_2 \\ &= S(T_1(v)) + S(T_2(v)) \text{ by linearity of } S \\ &= (S \circ T_1)(v) + (S \circ T_2)(v) \\ &= ((S \circ T_1) + (S \circ T_2))(v) \text{ by definition of } + \text{ in } \text{Hom}(V, U) \end{aligned}$$

$$\begin{aligned} \Rightarrow S \circ (T_1 + T_2) &= (S \circ T_1) + (S \circ T_2) \\ \text{Similarly } (T_1 + T_2) \circ R &= (T_1 \circ R) + (T_2 \circ R) \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \begin{array}{l} \text{These translate into} \\ \text{distributivity of matrix} \\ \text{multiplication over matrix } +. \end{array}$$

$$8) (a) \quad V \xrightarrow{T} W \xrightarrow{S} U$$

$$\begin{aligned} \text{Im}(T) &= T(V) \subset W \\ \therefore \text{Im}(ST) &= S(T(V)) \subset S(W) \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \begin{array}{l} \text{Subspace} \\ \text{Containments} \end{array}$$
$$\begin{array}{ccc} \dim \downarrow & & \downarrow \dim \\ \Rightarrow \text{rank}(ST) & \leq & \text{rank}(S) \end{array}$$

$$\text{Also } \text{rank}(ST) = \left(\text{rank of the map } T(V) \xrightarrow{S} W \right) \leq \dim(\text{domain})$$

$\downarrow$ $\text{rank}''(T)$

by dimension formula applied here

(b) each col of AB = lin combination of cols of A

$\Rightarrow$ col space of $AB \subset$ col space of A

$$\Rightarrow \text{rank}(AB) \leq \text{rank}(A)$$

each row of AB = lin combination of rows of B

$\Rightarrow$ row space of $AB \subset$ row space of A

$$\Rightarrow \text{rank}(AB) \leq \text{rank}(B) \quad (\text{row rank} = \text{col rank})$$

(c) Under bijection matrices $\longleftrightarrow$ lin maps, $\begin{array}{l} \text{multiplication} \longleftrightarrow \text{composition} \\ \text{column rank} \longleftrightarrow \text{rank} = \dim(\text{image}) \end{array}$

HW4 answers

5.1) intersection 0 and they span $\mathbb{R}^{n \times n}$ (how?)

5.2) "scalar matrices" $\{cI \mid c \in \mathbb{R}\}$

6.1) Eventually constant sequences. One nonexample $(a_1, a_2, \dots)$ where $a_i = i$

Chapter 1

4.3) Answer: $n+1$ (by induction)

4.6) (Sketch) Use our formula (complete expansion) to see that entries from B cannot contribute (due to the 0 block below A) and remaining terms can be grouped to get product $(\det A)(\det D)$. [Lengths of permutations in A part and D part add up.]

6.1) straightforward.

6.2) If $\bar{A}$ has integer entries, use $\det(A\bar{A}^T) = \det(A)\det(\bar{A}^T)$

If $\det(A) = \pm 1$, then $\pm$ cofactor/adj of A is $\bar{A}^T$ by Thm 1.6.9

M.7)

$(b-a)(c-a)(d-a)$

times this det

$$\begin{array}{c|c|c} \begin{array}{cccc} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{array} & \begin{array}{l} R_4 - aR_3 \\ R_3 - aR_2 \\ \rightarrow \\ R_2 - aR_1 \end{array} & \begin{array}{cccc} 1 & 1 & 1 & 1 \\ 0 & b-a & c-a & d-a \\ 0 & b^2-ab & c^2-ac & d^2-ad \\ 0 & b^3-ab^2 & c^3-ac^2 & d^3-ad^2 \end{array} \end{array} \rightarrow \begin{array}{cccc} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & b & c & d \\ 0 & b^2 & c^2 & d^2 \end{array}$$

It works better to use a^n to clear a^n , a^{n-1} to clear a^{n-1} , ..., a to clear a^2 , 1 to clear a than to use top 1 to clear all powers of a .

(c) Want $P(t) = a_0 + a_1 t + \dots + a_n t^n$ with $P(t_i) = b_i$ $i = 0, \dots, n$

This is equivalent to solving the following for $a_0, \dots, a_n$.

$$\begin{bmatrix} 1 & t_0 & \dots & t_0^n \\ \vdots & \vdots & & \vdots \\ 1 & t_n & \dots & t_n^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_n \end{bmatrix}$$

The coeff matrix is invertible by Vandermonde determinant. So there is a unique solution.

[Also see Artin example 1.2.22 and Lagrange interpolation formula]

See HW 2 #6

std basis vectors

8) Basis of domain $\mathbb{R}^5 = \underbrace{e_1, e_3}_{\text{std basis vectors}}$, basis vectors of $\text{Nul}(A)$

Basis of codomain $\mathbb{R}^3 = \text{cols 1 and 3 of } A, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ (or any } v \notin \text{col}(A))$